

## Visualize convolution

For  $x[n] = u[n] - u[n - 3]$  and

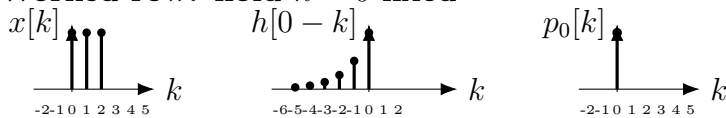
$$h[n] = \begin{cases} \left(\frac{1}{2}\right)^n, & 0 \leq n \leq 5, \\ 0, & \text{otherwise,} \end{cases}$$

For each fixed  $n$ , define  $p_n[k] = x[k]h[n - k]$ . The index  $k$  runs along every graph in a row;  $n$  stays fixed while completing that row.

1. Draw  $x[k]$ .
2. Flip and shift  $h[k]$  to draw  $h[n - k]$ .
3. Mark the overlap and draw  $p_n[k]$ .
4. Add the product samples:  $y[n] = \sum_k p_n[k]$ .

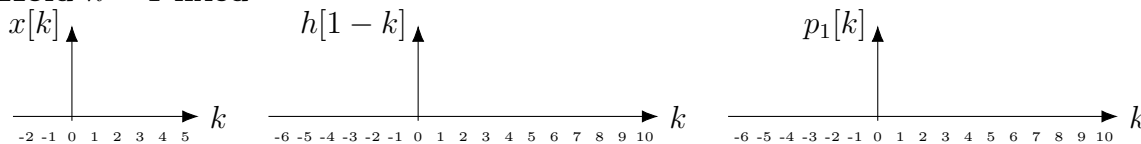
Use  $0 \leq k \leq 2$  for the support of  $x[k]$  and  $n - 5 \leq k \leq n$  for the support of  $h[n - k]$ . **At  $n = -1$ :** No overlap occurs, so  $x[k]h[-1 - k] = 0$  for all  $k$ , giving  $y[-1] = 0$ .

**Worked row: hold  $n = 0$  fixed**



Overlap index:  $k = 0$ . Hence  $y[0] = \sum_k p_0[k] = x[0]h[0] + x[1]h[-1] + x[2]h[-2] = 1$ .

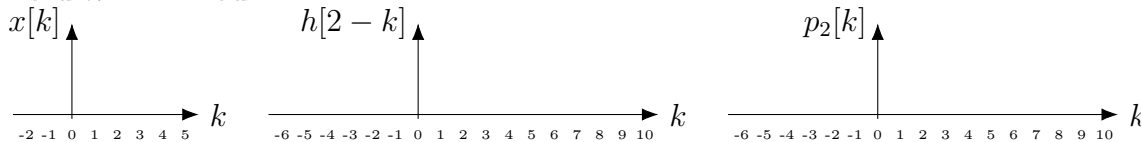
**Hold  $n = 1$  fixed**



Overlap indices:  $k =$  \_\_\_\_\_

$y[1] = \sum_k p_1[k] =$  \_\_\_\_\_

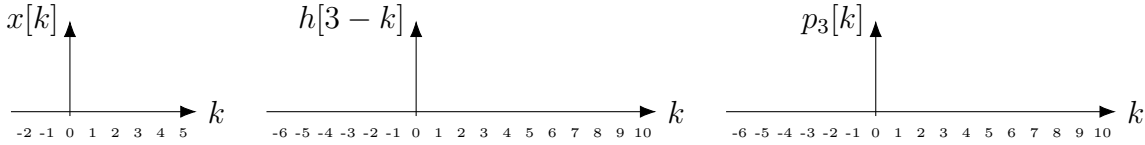
**Hold  $n = 2$  fixed**



Overlap indices:  $k =$  \_\_\_\_\_

$y[2] = \sum_k p_2[k] =$  \_\_\_\_\_

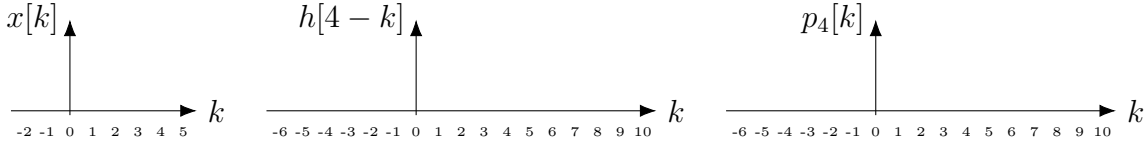
**Hold  $n = 3$  fixed**



Overlap indices:  $k =$  \_\_\_\_\_

$y[3] = \sum_k p_3[k] =$  \_\_\_\_\_

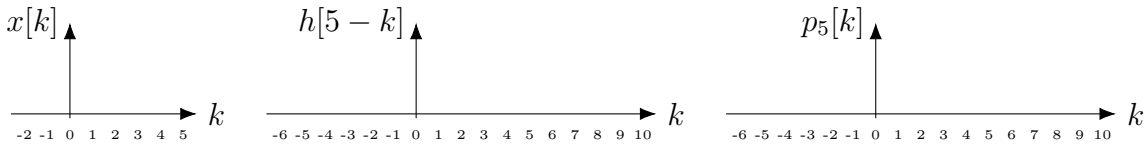
**Hold  $n = 4$  fixed**



Overlap indices:  $k =$  \_\_\_\_\_

$y[4] = \sum_k p_4[k] =$  \_\_\_\_\_

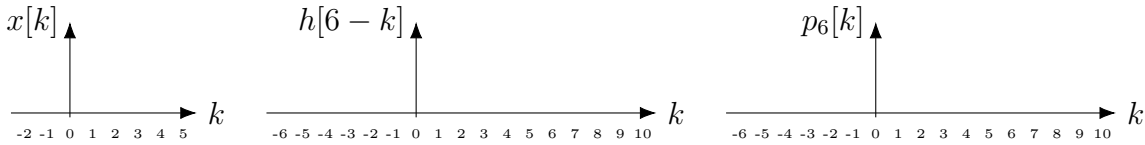
**Hold  $n = 5$  fixed**



Overlap indices:  $k =$  \_\_\_\_\_

$y[5] = \sum_k p_5[k] =$  \_\_\_\_\_

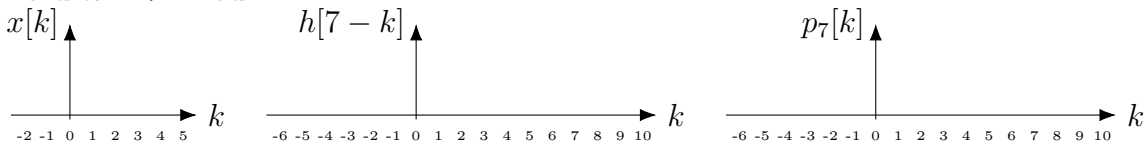
**Hold  $n = 6$  fixed**



Overlap indices:  $k =$  \_\_\_\_\_

$y[6] = \sum_k p_6[k] =$  \_\_\_\_\_

**Hold  $n = 7$  fixed**



Overlap indices:  $k =$  \_\_\_\_\_

$y[7] = \sum_k p_7[k] =$  \_\_\_\_\_

**Assemble the output**

Plot the samples from  $n = -1$  through  $n = 8$ , including the zero-valued samples. The nonzero support is \_\_\_\_\_  $\leq n \leq$  \_\_\_\_\_.

